

When you run *alien_invasion.py* now, the ship should stop moving at either edge of the screen. This is pretty cool; all we've done is add a conditional test in an if statement, but it feels like the ship hits a wall or force field at either edge of the screen!

Refactoring `_check_events()`

The `_check_events()` method will increase in length as we continue to develop the game, so let's break `_check_events()` into two separate methods: one that handles `KEYDOWN` events and another that handles `KEYUP` events:

```
alien_invasion.py    def _check_events(self):
                    """Respond to keypresses and mouse events."""
                    for event in pygame.event.get():
                        if event.type == pygame.QUIT:
                            sys.exit()
                        elif event.type == pygame.KEYDOWN:
                            self._check_keydown_events(event)
                        elif event.type == pygame.KEYUP:
                            self._check_keyup_events(event)

                    def _check_keydown_events(self, event):
                        """Respond to keypresses."""
                        if event.key == pygame.K_RIGHT:
                            self.ship.moving_right = True
                        elif event.key == pygame.K_LEFT:
                            self.ship.moving_left = True

                    def _check_keyup_events(self, event):
                        """Respond to key releases."""
                        if event.key == pygame.K_RIGHT:
                            self.ship.moving_right = False
                        elif event.key == pygame.K_LEFT:
                            self.ship.moving_left = False
```

We make two new helper methods: `_check_keydown_events()` and `_check_keyup_events()`. Each needs a `self` parameter and an `event` parameter. The bodies of these two methods are copied from `_check_events()`, and we've replaced the old code with calls to the new methods. The `_check_events()` method is simpler now with this cleaner code structure, which will make it easier to develop further responses to player input.

Pressing Q to Quit

Now that we're responding to keypresses efficiently, we can add another way to quit the game. It gets tedious to click the X at the top of the game window to end the game every time you test a new feature, so we'll add a keyboard shortcut to end the game when the player presses Q:

```
alien_invasion.py    def _check_keydown_events(self, event):
                    --snip--
                    elif event.key == pygame.K_LEFT:
```
